Electronic Intelligence (ELINT)

Quick briefing

Electronic intelligence (ELINT) interprets non-communication emissions such as radar signals. Distinguish the features you observe, what they suggest about the emitter, and what evidence would support a firmer assessment.

Know the facts

ELINT and COMINT · ELINT examines non-communication emissions and their technical signatures. COMINT examines communications such as voice, text or data messages.

Electronic Support and ELINT · Electronic Support serves a commander's immediate awareness needs. ELINT produces assessed technical intelligence that can build or update emitter knowledge.

Electronic Order of Battle · The lesson distinguishes Friendly, Adversary and Neutral EOBs. ELINT contributes to building and updating the Adversary EOB.

Radar echo · A pulsed radar uses the elapsed time for a transmitted pulse to return as an echo to estimate range. A passive receiver observes a signal on its outbound journey.

Signature parameters · Carrier frequency, pulse width, pulse repetition interval, scan behavior and other observed parameters help describe a radar signature. PRF is the reciprocal of PRI in seconds.

Search and track · A rotating or sector sweep is consistent with search. A more focused pattern with faster pulse repetition can suggest a shift toward tracking.

Understand the reasoning

A behavior change can justify a prompt update while identification remains tentative. Report what changed and state how strongly it supports the interpretation.

One common parameter is a weak match. Look for agreement across several signature features and preserve uncertainty when a library entry is incomplete.

Philippine Army classroom exercise

A Philippine Army classroom team compares two simulated radar records. The same exercise emitter changes from a repeating sweep to a more focused pattern with faster pulses. A library entry matches its carrier band but lists a different pulse width.

Decision · What should the team report about behavior and identity?

Worked analysis · Report the change as consistent with a shift from search toward tracking, and retain the recorded features that support that interpretation. Keep the equipment identity provisional because the band match alone is weak and another feature differs. The behavior update can reach the instructor before the identity is resolved.

How to practice

1 · Use the supplied synthetic records to list measured features for each observation.

2 · Compare the scan behavior and pulse timing across the records and describe the observed change.

3 · Compare the set of features with the exercise signature library and mark matches, conflicts and unknowns.

4 · Write an exercise update with separate observation, tentative assessment and validation fields.

Check yourself

Which SIGINT sub-discipline studies a radar emission's technical signature? · Electronic intelligence (ELINT).

If the pulse repetition interval is 2 milliseconds, what is the pulse repetition frequency? · 500 hertz, because 1 divided by 0.002 seconds is 500 pulses per second.

Course source

Electronic Intelligence (ELINT)